Tkinter Components and Widgets

Tkinter provides many components, called widgets, that help build interactive graphical applications. A program can contain multiple visual elements such as windows, frames, buttons, and text fields. These widgets allow users to enter data, choose options, or perform actions. Each widget has its own purpose, and combining them creates a complete interface.
Tkinter also supports layout management, so widgets can be arranged neatly on the screen.



CLASS ACTIVITY

Create a main window and divide it into two frames. Add different widgets in each frame to organize layout.

Creating Windows and Adding Frames

- A Tkinter program begins by creating a main window, which serves as the base for all other elements.
    - The window can be resized, given a title, and designed to hold different widgets.
- **Frames** are sections within the window that help organize content.
    - Frames divide the window into smaller parts, making it easier to group related widgets. This structure improves clarity and keeps the interface well-arranged.

Example:

Code:

```
import tkinter as tk

# Create main window
window = tk.Tk()
window.title("Tkinter Frames Example")
window.geometry("400x300")

# Create top frame
top_frame = tk.Frame(window, bg="lightblue", height=100)
top_frame.pack(fill="x")

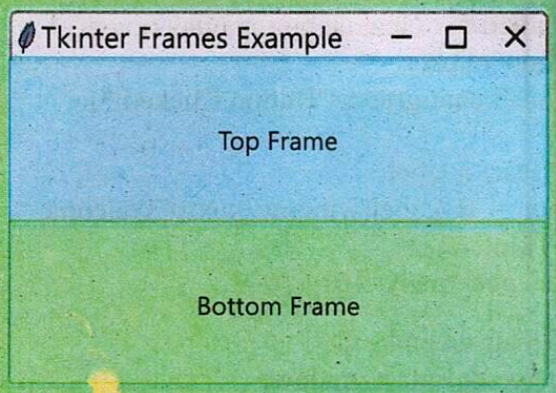
# Create bottom frame
bottom_frame = tk.Frame(window, bg="lightgreen", height=200)
bottom_frame.pack(fill="both", expand=True)

# Add widgets to top frame
label_top = tk.Label(top_frame, text="Top Frame", bg="lightblue")
label_top.pack(pady=20)

# Add widgets to bottom frame
label_bottom = tk.Label(bottom_frame, text="Bottom Frame", bg="lightgreen")
label_bottom.pack(pady=50)

# Run the application
window.mainloop()
```

Output:



- Tk( ) → creates main window
- Frame → divides window into sections
- top_frame and bottom_frame → organize layout
- pack() → places frames and widgets
- Frames help keep the GUI clean and well-structured

Common Widgets: Labels, Buttons, Entry Fields, Menus, and List boxes

Widgets are the basic elements used to build a graphical user interface.

Label: A Label is used to display text or messages on the screen.
Button: A Button allows the user to perform an action when it is clicked.
Entry field: An Entry field is used to enter text, such as a name or password.
Menu: A Menu provides a list of options at the top of the window.
Listbox: A Listbox shows a list of items and allows selection from the list.

Widget	Purpose
Label	Displays text or messages
Button	Performs an action when clicked
Entry	Accepts single-line text input (e.g., name, password)
Menu	Provides a list of options at the top of the window
Listbox	Shows a list of items and allows selection
Frame	Divides the window into sections to organise widgets